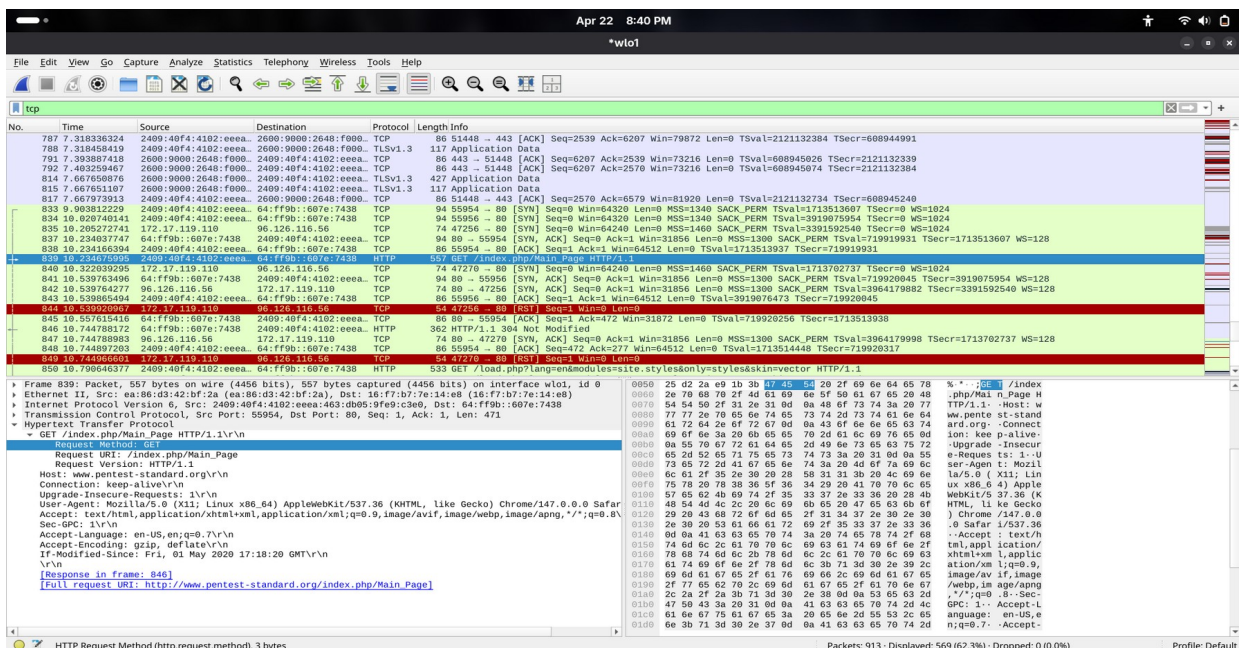


Day 2 - Networking & Protocols for Pentesters

HTTP REQUEST

- Website used: http://www.pentest-standard.org/index.php/Main_Page
- Filter used: tcp
- HTTP GET request at Packet No. 839
- 3 way-handshake at Packet No. 835 (SYN), Packet No. 837 (SYN-ACK), Packet No. 838 (ACK)
- In Packet Details Pane, using HTTP, the browsing details are shown in human readable form.
- In Packet Bytes Pane, it shows the User Agent, Host, OS, etc in clear text ASCII
- TCP protocols mentioned (light blue) uses IPv6, but the HTTP protocol with TCP (light green) uses IPv4, which shows up basic configs in firewall.
- This is why HTTP is vulnerable and ease for credential sniffing



HTTPS REQUEST

- Website used: <https://www.geeksforgeeks.org/>
- Filter used: tls
- HTTPS GET at Packet No. 270, mentioned as Application Data
- The payload is encrypted, only cipher text is visible in Packet Details Pane in Encrypted Application Data
- The corresponding HEX and ASCII is shown in the Packet Bytes Pane
- 3 way-handshake: Packet No. 257 (Client Hello), Packet No. 266 (Server Hello)
- Every packets are mentioned in IPv6
- This makes HTTPS to be secured as the payloads are encrypted.
- Though encrypted, the metadata are visible, makes to identify what service are running.

Apr 22 10:01 PM

https.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tls

No.	Time	Source	Destination	Protocol	Length	Info
22	13.963831424	2409:40f4:4102:eeee...	2404:6800:4002:803:...	QUIC	1292	Initial, DCID=c96d7832ac2e3ee, PKN: 3, PING, PADDING, CRYPTO, PING, PING, CRYPTO, PING, PING, PADDING, CRYPTO,...
28	14.081855427	2404:6800:4002:803:...	2409:40f4:4102:eeee...	QUIC	1292	Initial, SCID=e99d7832ac2e3ee, PKN: 5, CRYPTO, PADDING
60	19.448079780	2409:40f4:4102:eeee...	2600:9000:2579:6c00...	QUIC	1292	Initial, DCID=9ce99357beeb09f2, PKN: 2, PING, CRYPTO, CRYPTO, CRYPTO, CRYPTO, CRYPTO, PING, PADDING, PING, CRYPTO, PING, CRYPTO, PADDI...
62	19.603642682	2600:9000:2579:6c00...	2409:40f4:4102:eeee...	QUIC	1262	Initial, SCID=8d260681db2031b698c78d99f985fae988223ee8, PKN: 1, ACK, CRYPTO
63	19.603642903	2600:9000:2579:6c00...	2409:40f4:4102:eeee...	QUIC	1262	Handshake, SCID=8d260681db2031b698c78d99f985fae988223ee8
251	25.262948639	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	858	Client Hello (SNI=imgs.search.brave.com)
253	25.307258987	2409:40f4:4102:eeee...	2600:9000:2579:6c00...	QUIC	1292	Initial, DCID=80bf5264abb0dc7, PKN: 2, CRYPTO, PING, CRYPTO, PADDING, PING, PING, PADDING, PING, PADDING, CRYPTO, PADDING, PING, PADD...
257	25.308876363	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	705	Client Hello (SNI=imgs.search.brave.com)
266	25.322629266	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	200	Server Hello, Change Cipher Spec, Application Data, Application Data
268	25.323001062	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	150	Change Cipher Spec, Application Data
270	25.323255847	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	178	Application Data
271	25.323513731	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	1134	Application Data
272	25.323547033	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	1156	Application Data, Application Data, Application Data
273	25.323659096	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	509	Application Data, Application Data
280	25.438934625	2600:9000:2579:6c00...	2409:40f4:4102:eeee...	QUIC	1262	Initial, SCID=8d079501dbfc4fc7d0b4431ad3e6f24e0ea5dc82, PKN: 1, ACK, CRYPTO
281	25.438934746	2600:9000:2579:6c00...	2409:40f4:4102:eeee...	QUIC	1262	Handshake, SCID=8d079501dbfc4fc7d0b4431ad3e6f24e0ea5dc82
292	25.431208939	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	265	Application Data
293	25.431209049	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	148	Application Data
300	25.432444550	2409:40f4:4102:eeee...	2600:9000:257a:1e00...	TLSv1.3	117	Application Data
301	25.433091651	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	117	Application Data
303	25.433224610	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	1841	Application Data
306	25.433224991	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	1604	Application Data
310	25.433356910	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	1294	Application Data
311	25.433373759	2600:9000:257a:1e00...	2409:40f4:4102:eeee...	TLSv1.3	1294	Application Data

Frame 270: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface wlo1, id 0

Ethernet II, Src: ea:86:d3:42:bf:2a (ea:86:d3:42:bf:2a), Dst: 16:f7:b7:7e:14:e8 (16:f7:b7:7e:14:e8)

Internet Protocol Version 6, Src: 2409:40f4:4102:eeee:463:db05:9fe9:c3e0, Dst: 2600:9000:257a:1e00:7:7f8e:6d00:93a1

Transmission Control Protocol, Src Port: 33490, Dst Port: 443, Seq: 2045, Ack: 1323, Len: 92

Transport Layer Security

[Stream index: 0]

TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol

Opaque Type: Application Data (23)

Version: TLS 1.2 (0x0303)

Length: 87

Decrypted Application Data: 4475a803e5f68d528495c69d58e25b7d327159611d434b05774657623d5f73718a045041a87fa7ba938

[Application Data Protocol: Hypertext Transfer Protocol]

0000 16 f7 b7 7e 14 e8 ea 86 d3 42 bf 2a 86 dd 60 09B....
0010 97 a5 00 7c 06 40 24 09 40 f4 41 02 ee ea 04 63 ...|(\$ @A...c
0020 db 05 9f e9 c3 e0 26 00 90 00 25 7a 1e 00 00 07&...Sz...
0030 7f 8e 6d 00 93 a1 82 d2 01 b0 f0 3d cc 52 05 46 ..B.....R.F
0040 1a e0 80 18 00 42 e6 95 00 00 01 01 08 0a b2 edB.....
0050 06 7e 33 f7 36 9c 17 03 03 00 57 84 75 a0 03 05 ...-3 6...wpu...
0060 0f 69 52 6a 85 c0 00 38 62 50 70 82 71 69 61 d0x...[]?aYq
0070 43 4b 05 77 46 57 62 3d 5f 73 71 8a 04 50 41 a8 CK:wPwDs...:PA
0080 7f a7 ba 93 09 d4 ad df 9c b9 2e 45 b7 4b 2f d6:..E./K/
0090 07 23 2a 12 e4 6d 17 f2 94 4f d8 77 f7 f6 00:..0-W-W-
00a0 4e 34 24 8f 7d c8 05 e2 cd 3c 63 c8 0c 73 b7 d4 nGS...-cS...s
00b0 88 1e

Payload is encrypted application data (tls.app_data), 87 bytes

Packets: 575 - Displayed: 85 (14.8%) - Dropped: 0 (0.0%)

Profile: Default